NETFLIX CONTENT TRENDS

Exploratory Data Analysis Using Python

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Tools: Python, Pandas, NumPy, Matplotlib, Jupyter Notebook

1. Introduction

Streaming platforms organise large and diverse catalogues. This project examines the composition of a Netflix titles dataset to identify patterns across content types, categories, countries, catalogue additions and programme duration. The goal is to turn raw metadata into clear, interpretable observations, rather than infer audience viewing behaviour from a titles list.

2. Dataset and analytical approach

The Kaggle Netflix Movies and TV Shows dataset contains **8,807 titles and 12 original columns**. The analysis checked duplicate show identifiers and missing fields, parsed addition dates, separated movie running times from TV show seasons, and summarised counts by content type, first-listed genre and first-listed country. Multi-valued genre and country fields were simplified to their first entry for mutually exclusive comparisons.

3. Data quality

Missing metadata includes **2,634 director values, 825 cast values, 831 country values and 10 addition dates**. These gaps were not replaced with guesses. Counts are based on the supplied historical dataset and should not be mistaken for the live Netflix catalogue.

4. Main findings

Catalogue mix: The dataset includes **6,131 movies (69.6%)** and **2,676 TV shows (30.4%)**. This is a title-count comparison, not a comparison of viewing time or popularity.

Primary categories: Dramas is the most frequent first-listed category, with 1,600 titles. Because individual titles can carry multiple category labels, the result depends on the first-label convention.

Production geography: United States is the most common first-listed country (3,211 titles). Co-productions may involve countries beyond the first listed.

Catalogue additions: 2019 has the highest number of recorded additions (2,016 titles). This reflects dataset addition dates, not releases or audience demand.

Duration: The median recorded movie duration is 98 minutes; the median recorded TV show length is 1 season(s). These represent different units and are not directly comparable.

5. Visual analysis — content mix and categories

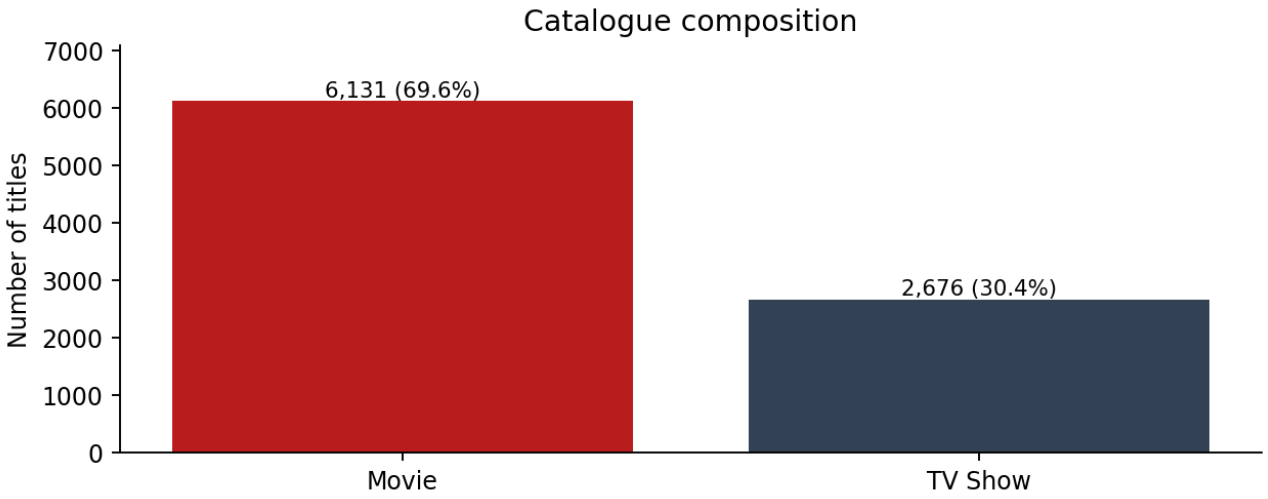


Figure 1. Movies make up more than two-thirds of the titles in this snapshot.

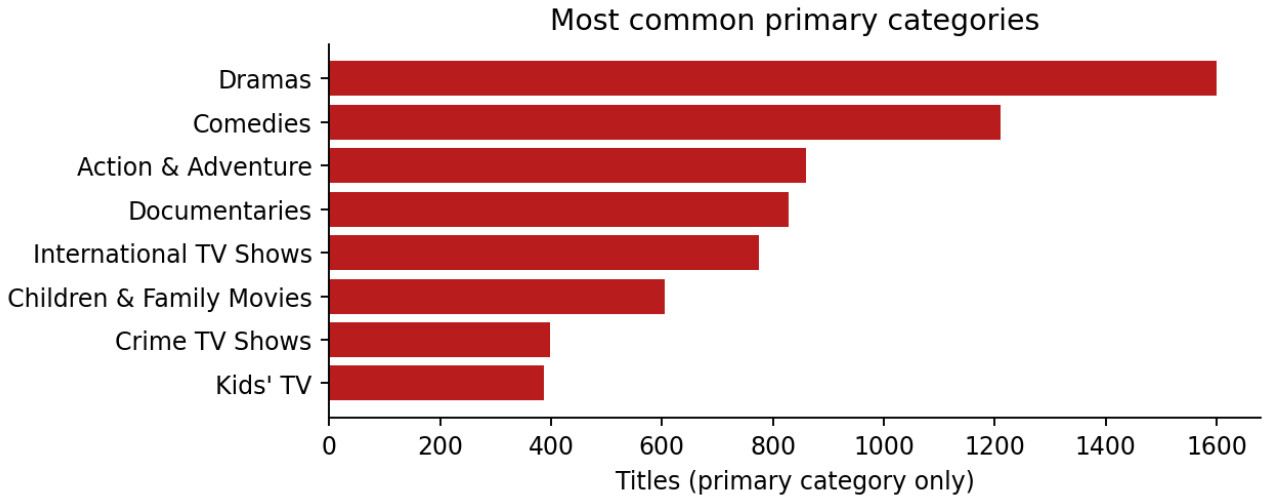


Figure 2. Top first-listed categories; each title is counted in one category only.

6. Visual analysis — countries and time

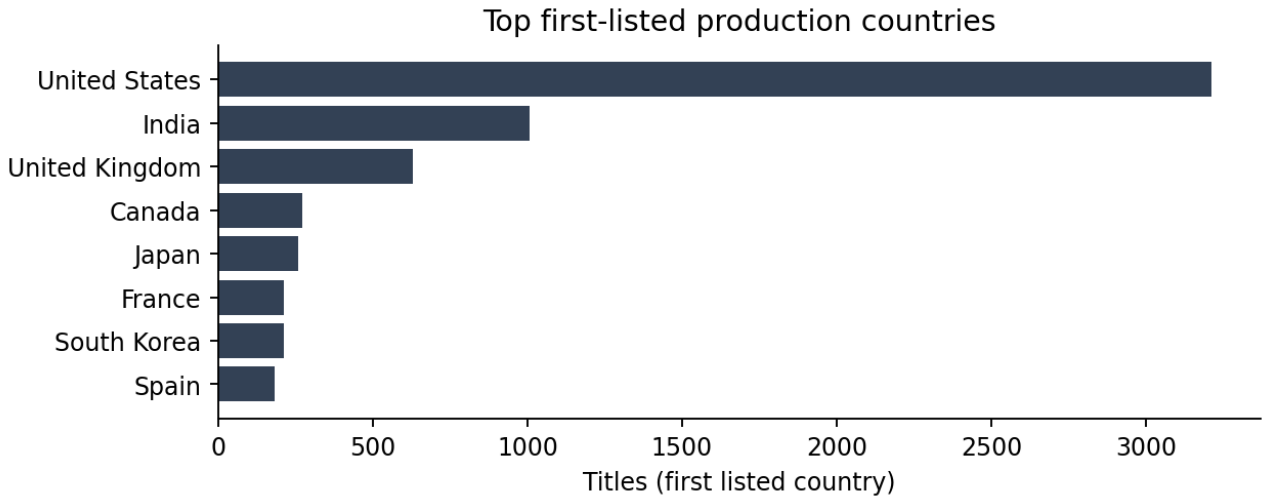


Figure 3. Country counts use the first country listed, excluding unknown entries.

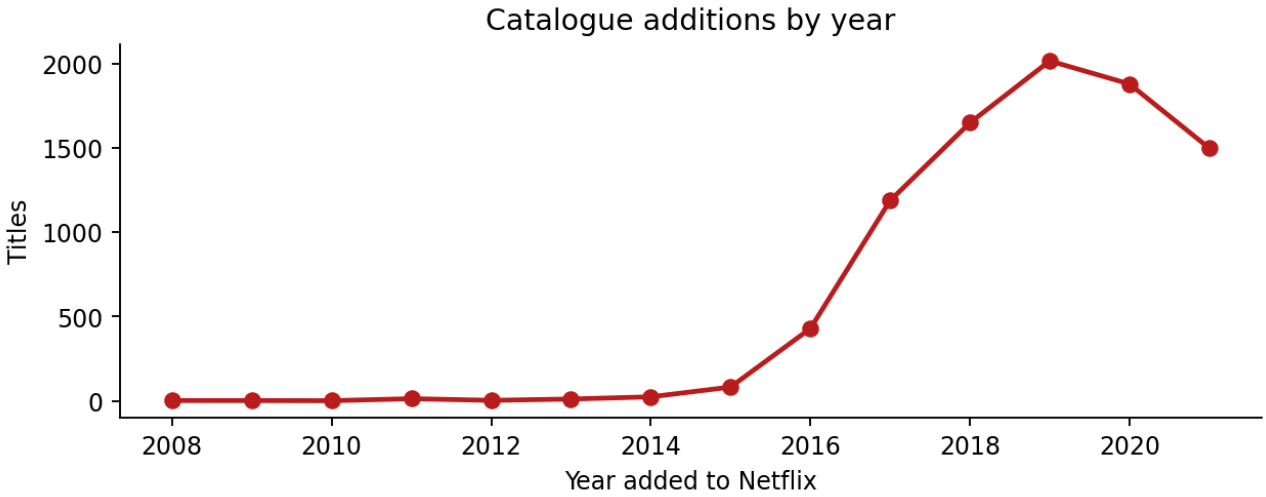


Figure 4. Year added is different from original release year; recent-year coverage is incomplete.

7. Visual analysis — movie duration

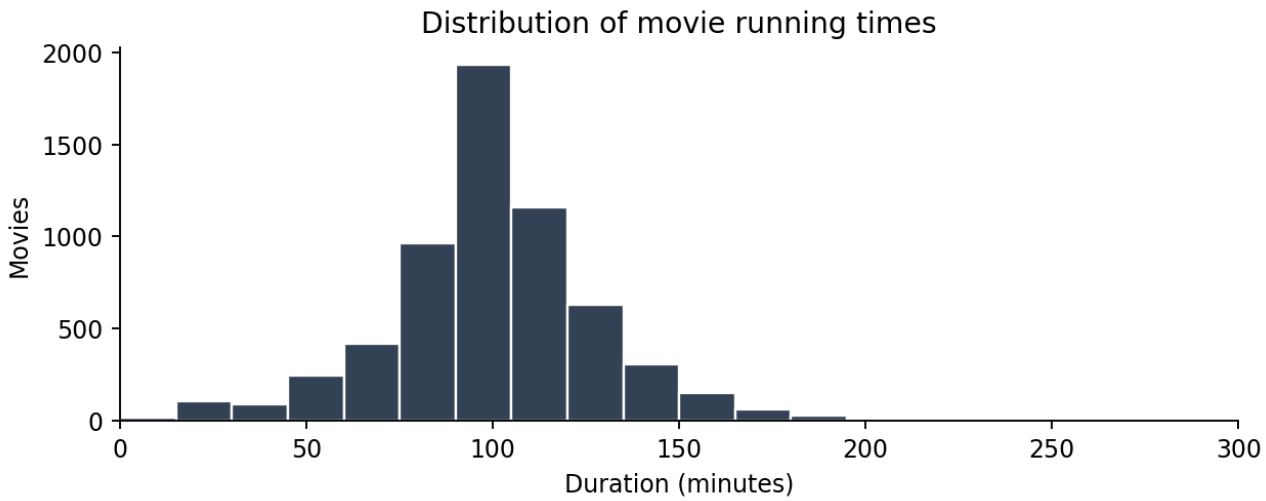


Figure 5. Distribution of recorded movie running times; extreme durations are outside the displayed 0–300-minute range.

8. Interpretation and recommendations

The catalogue is movie-heavy, while the country and category breakdowns show an uneven distribution of titles. For a content planning exercise, these results could help identify where the catalogue is concentrated and where further research might be useful. However, decisions about investment, licensing or recommendations would also require viewing figures, audience demographics, licensing costs and up-to-date availability.

9. Limitations

This is a historical catalogue snapshot, not a current inventory or viewership dataset. Some records have missing metadata. The first-listed-country and first-listed-category rules simplify multi-label information, and counts cannot establish popularity, revenue or causation. The recorded year of addition is not the original release year.

10. Conclusion

Using Python, this analysis transformed Netflix title metadata into five visual summaries and interpretable findings. The exercise strengthened practical skills in cleaning imperfect data, creating grouped summaries and communicating results carefully. The strongest conclusion is about the composition of the recorded catalogue—not about how audiences watched it.

Dataset reference

Shivam Bansal. *Netflix Movies and TV Shows*. Kaggle. https://www.kaggle.com/datasets/shivamb/netflix-shows